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You can't be neutral on a moving train

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Abstract

The commentary argues the authors employ misdirection and strawmanning to cast others as polarized extremes and themselves as the reasonable centrists. We argue that these patterns of misrepresentation ultimately damage any consensus and middle ground they claim to hope to reach.

Futrell & Mahowald (F&M) want readers to see them as reasonable centrists, but do so by casting other positions as polarized extremes via misdirection and strawmen. Their centrism is an illusion and comes at the cost of saying anything substantive or advancing the conversation. By the end, the sheep becomes a wolf: Claudette is inviting Norm for dinner, and Norm is on the menu. With an invitation like that, no wonder linguists would rather continue to focus on the human capacity for language in all its glorious variations.

F&M frame radical positions as a middle ground by strawmanning the debate. For example, regarding learnability constraints, they state *"There is no logical necessity for a restrictive formalism to explain language learning."* and *"The key mistake was . . . that inductive bias has to come from restrictions on the learner's hypothesis space."* They highlight Wilson's (2025) position paper describing several deep learning phenomena using the PAC-Bayes framework via soft' biases. In reality, these are a set of hidden "hard" biases: a fixed computable hypothesis class, a fixed description-length prior tied to the architecture, I.I.D. sampling assumptions, bounded-loss assumptions, task-dependent representational primitives (by fallaciously appealing to Li and Vitanyi's (2008) "additive-constant" invariance theorem), and other capacity constraints. While description-length priors are coherent, Wolpert (2021) shows a bias only justifies a particular hypothesis class to the extent it matches the (unknown) distribution over tasks (see James et al., 2023 and Curth, Jeffares, & van der Schaar, 2023 on task-dependent complexity and double-descent). Without explicitly formalizing that alignment, moving from "real-world language data is inherently compressible" to endorsing biases like those above is meaningless. Whether hypotheses are restricted, admissible data presentations are restricted, or learning resources in time and space are unrestricted, there are fundamental tradeoffs in solving learning problems (Angluin, 1988; Goyal & Bengio, 2022, a.o.).

F&M's own candidate constraints like sensitivity and information locality are also model-dependent (Hahn & Rofin, 2024; Someya et al., 2025), task-dependent (Hahn, Jurafsky, & Futrell, 2021), and ultimately "hard." Even Wilson and colleagues agree that *"it is largely the design of the architecture that makes generalizable solutions more easily accessible"* (Goldblum et al., 2024). Like all "domain-general" biases, F&M's candidates turn out to be just "domain-specific" biases of a different sort (Rawski & Heinz, 2019). In other words, the domains and biases F&M want are real, and the others are not. F&M can only achieve centrism from such an extreme position by misassigning people to trivially false claims, like that Rawski and Baumont (2023)'s 1-page paper says Large Language Models (LLMs) *"can learn to approximate anything"* when it does not. Strawmanning debates by mischaracterizing one perspective to position oneself as a middle ground, while ignoring the substance of the issues, is disingenuous and ultimately leads nowhere.

F&M cover these realities with doublespeak and fallacious arguments. They hype LLM success, concede the successes are disagreed upon and may always emerge from shallow heuristics (Liu et al., 2022; McCoy, Pavlick, & Linzen, 2019; Petty et al., 2025; Vafa et al., 2025; Vázquez Martínez et al., 2026, a.o.), backtrack to *"this is not the right place to adjudicate these issues,"* and then reassert that LLMs have *"learned 'the real thing'"* and are *"not purely relying on heuristics or memorization or cheap tricks."* Even their interpretability "thought experiment" about an idealized "c-command neuron" is a revamp of standard cognitive neuroscience techniques with the same reification fallacies and circularity (Buzsaki, 2019). Again they admit *"LMs just don't represent anything in such human-interpretable ways,"* yet insist they are *"linguistically interesting."* This is not a serious discussion.

Beyond the banal observation that *"there are increasingly researchers using language models to ask questions that are informed by and which can inform linguistic theory,"* what is F&M's actual point? Simultaneously insisting on the Eliminativist fallacy (Smolensky, 1988) that restrictive symbolic formalisms are *"dogma,"* while claiming *"we are not eliminativists about grammatical structure,"* is incoherent. Taking a Limitivist view (Smolensky, 1988) where symbolic formalisms are an "approximate explanation" and *"graded, probabilistic, and fuzzy"* subsymbolic primitives are cognitively real, they bizarrely omit 30 years of neural networks in generative linguistics via Optimality Theory and Harmonic Grammar (Prince & Smolensky, 1997; Smolensky & Legendre, 2006). Advocating for "domain-general" learning biases, which end up being domain-specific, paints them into a Dennettian corner to conclude the "troll's truism" (Schackel, 2005) that *"linguistic structure is real."* Little else remains, certainly nothing new.

Here we see the tragedy of the centrist, ruled by *"the zeitgeist"* in F&M's terms, at the cost of ideas or content. Like most self-described centrists, however, their strategies serve to cover another goal. F&M seem to want linguistics to effectively become a niche within psychology (Núñez et al., 2019), where replicability and theory crises (Eronen & Bringmann, 2021) mean *"theories rise and decline, come and go, more as a function of baffled boredom than anything else; and the enterprise shows a disturbing absence of that cumulative character that is so impressive"* (Meehl, 1978). Linguistic theory is so robust and successful at phenomena discovery that LLM performance benchmarks (Gauthier et al., 2020, a.o.) and evaluations (Hale & Stanojević, 2024, a.o.) crucially rely

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on them. When neural models are used to trivialize the substance of linguistic theory, as F&M do with competence and performance (Berwick, 2018; Firestone, 2020; West et al., 2024), linguists have no reason to engage. Replacing theory-driven science with model-driven science, and phenomena with effects-chasing, is a profound retreat from the core insights which gained linguistics membership in cognitive science in the first place.

F&M wonder why many linguists remain uninterested in LLMs, which have not yet discovered virtually any new phenomena nor explained anything of theoretical significance. Yes, LLMs are a technological advance and engineering achievement. Nonetheless, language models, in whatever guise, will continue to be a scientific embarrassment to language scientists as long as they do not offer robust phenomena discovery, counterfactual support, explanatory depth, and other scientific insight into the fundamental properties of the human capacity for language.

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LLMs are not children: They have to earn our love

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